

A!

TILA · DAHA WORKSHOP · FLASH TALK

Rethinking research computing and data infrastructures for non-human users

AI agents as the new users of research computing and data services.

Enrico Glerean, Staff Scientist, Aalto University · 7 October 2026

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What is AI?

What is AI in research? Topic, method, tool

Reference: [Best practices for AI use in research work \(Aalto.fi\)](#)

AI as a topic

AI is what is being studied.

- A new ML model or architecture
- A new algorithm for training
- How people interact with AI chatbots

AI as a method

AI is how the data are analysed.

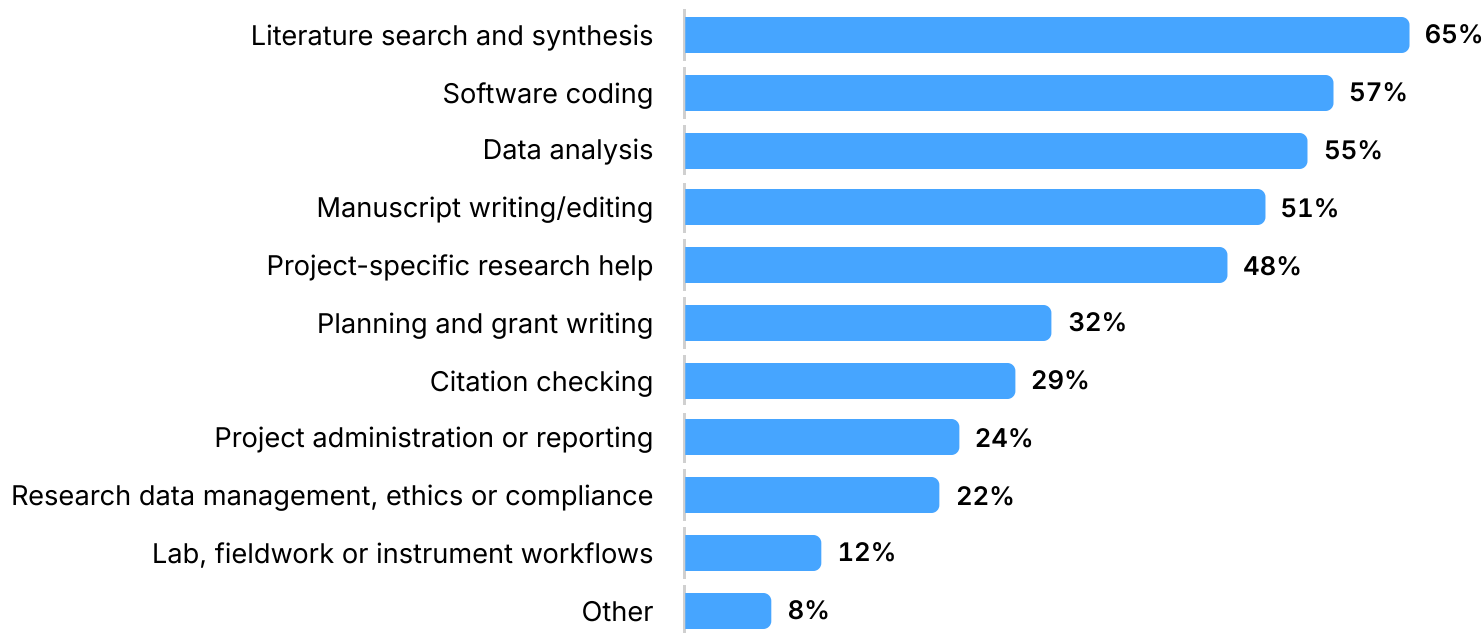
- A deep neural network that classifies tissue in cancer imaging data
- scikit-learn models in the analysis of a dataset

AI as a tool

AI is something that helps with the work.

- Brainstorming with ChatGPT
- Claude Code writing analysis scripts
- An AI assistant helping to fill in a data management plan

What do researchers want AI help with?



Share of respondents per task, multiple choices per respondent. Source: Aalto University survey on AI agents, 2026 (N = 183).

The tool became a user

1. **Chat window:** copy and paste
2. **Coding assistant:** in the editor
3. **Agent:** logs in, reads files, submits jobs

Next, the fourth role: AI as a *user* acting for a researcher.

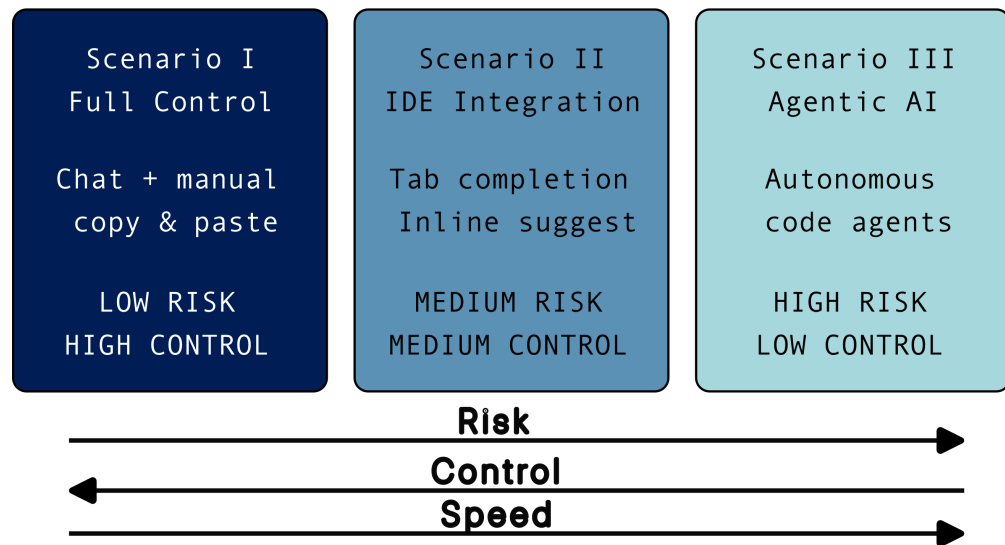


Figure: [CodeRefinery](#), [Coding with AI](#)

What is an AI agent?

An AI agent pairs the model with more components so that it can **take real actions**.

- **The model**, the "intelligence box": the LLM itself.
- **The scaffolding**, between the two: system prompts, tool descriptions, and the rules for how context is assembled and formatted before it reaches the model.
- **The harness**, the "body": the code around the model that calls it, carries out whatever it asks for, and decides when the task is done.

Harness · the body

calls the model · runs its actions · decides when done

Scaffolding

system prompt · tool descriptions · context assembly

Model

the LLM · the intelligence box



Environment

files

shell

Slurm

APIs

web

Adapted from Hugging Face, ["Harness, Scaffold, and the AI Agent Terms Worth Getting Right"](#), 2026

Machines were clients, agents become users

Today

APIs

scripts

workflows

CI

schedulers

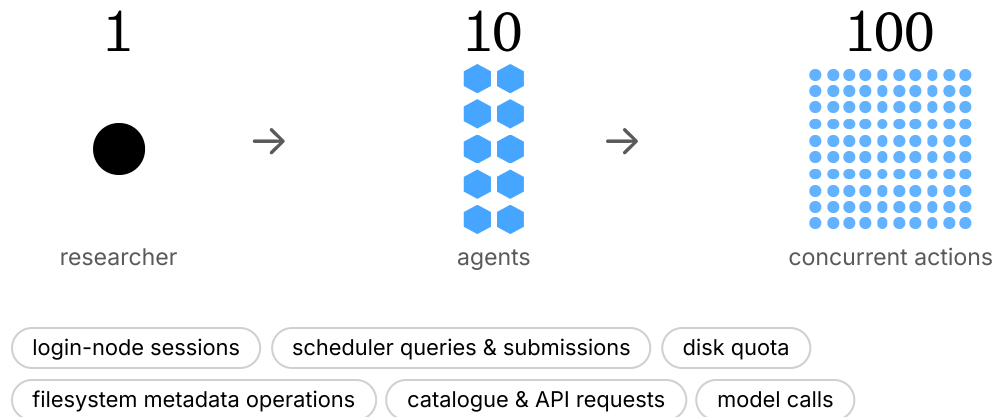
...already query machines.

Agents

- persistent
- adaptive
- retrying
- parallel
- capable of spawning more activity

The novelty is not machine access; it is that machines can now **act, adapt, and repeat at a scale humans cannot.**

One researcher $\neq$ one workload



THE OLD ASSUMPTION

1 account $\approx$ 1 human $\approx$ human-speed interaction

THE 2030 ASSUMPTION

1 researcher $\rightarrow$ N agents $\rightarrow$ N $\times$ M concurrent actions

A service designed for 5,000 users faces activity from 500,000 concurrent actions.
What will break first?

New opportunities for 2030

Compute automates all research tasks

LLM inference for AI agents will be the **biggest use of compute**. Compute is not just data analysis; it automates everything. Researchers become **managers and architects** rather than chiselling code by hand.

European AI

Let's stop sending European data (and money) to the USA: **local AI models** are the only way to go.

Local inference for all researchers and EU citizens.

FAIR for AI agents

FAIR has to speak the language of agentic AI: **MCP servers**, `SKILL.md`, `llms.txt`, machine-readable docs.

Diversity of interfaces will mean challenges for interoperability.

CLOSING QUESTION

How do we design for 2030 when the technology changes every six months?

Planning for **containment, interoperability, and sovereignty**
rather than trying to predict the next tool or use case.

Thank you! – Enrico Glerean

EXTRA SLIDES

Bonus materials

The things we won't be doing by hand in 2030

PLAN

Filling in the data management plan, privacy notice, consent forms

COLLECT

Writing the data access application, APIs, user interfaces

ANALYSE

Writing HPC scripts, watching the queue, debugging and resubmitting

STORE

Moving data between scratch and archive. Git push.

PUBLISH

Writing metadata, README, documentation, web pages

THROUGHOUT

Emailing the service desk, reading the docs

The agent does the planning, typing, coding, archiving and publishing. **The researcher should still decide, verify, and approve.** What could possibly go wrong?

RISKS

The risks

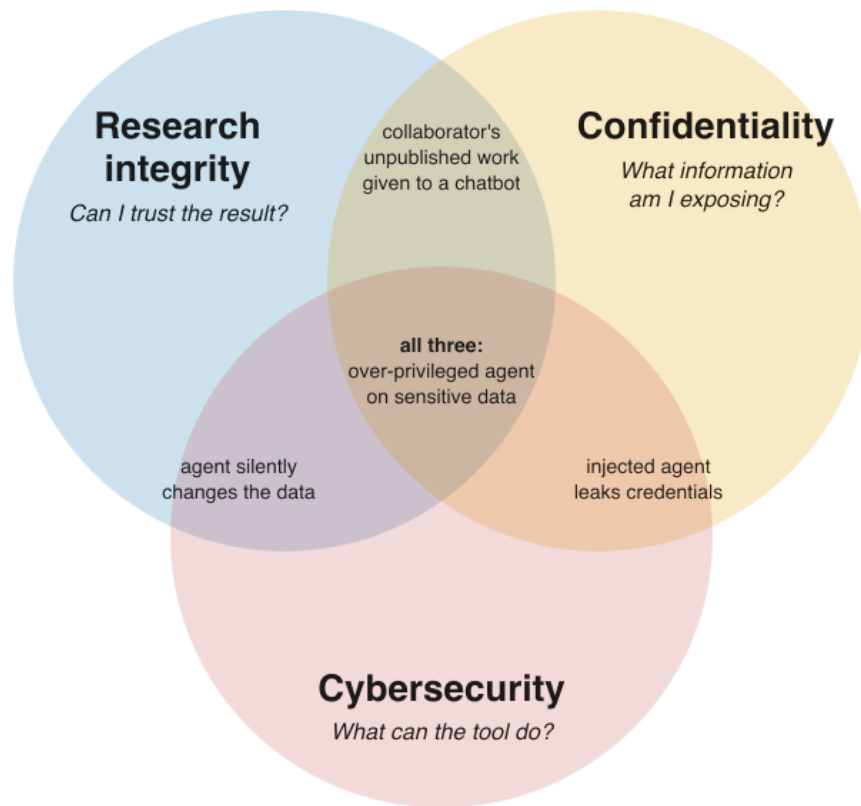
Research integrity: code that runs but is *plausibly wrong*, silent data loss, undisclosed AI use.

Confidentiality: unpublished work, participant data and credentials sent to third-party providers.

Cybersecurity: prompt injection, hallucinated or typosquatted packages, insecure generated code, agents with too much access.

Validate. Minimize. Contain.

Source: [CodeRefinery, Coding with AI: Security](#)



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Background in neuroimaging. I train and support researchers with:

- Personal data: anonymisation, secure computing
- Medical images, clinical trials
- Research ethics and integrity (AI and new technologies)
- Statistics, open science

I coordinate **LUMI AI Factory** trainings at Aalto University.

Other affiliations / COI

- **CodeRefinery**: Nordic network teaching computational reproducibility
- **Finnish Reproducibility Network**: national network raising awareness of reproducibility
- **Support Pool of Experts, European Data Protection Board**: open training materials on personal data, AI and cybersecurity
- Member of the **EU AI Act Advisory Forum**
- Teacher in the **Data Steward training programme**, Tampere University